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unit 5 Knoweldge Check STEAM Questions & Critical Thinking

PDF Documents

PDF Document 1: 1731845620523-Chapter 7_ unit5 Knowledge Check.pdf

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KNOWLEDGE CHECK

1 What does the range $5\text{cm} < \text{working range} < 15\text{cm}$ for a smart fan indicate?

- ☐ A The fan works only exactly at 5cm or 15cm.
- ☐ B The fan operates when an object is more than 15cm away.
- ☐ C The fan operates when an object is between 5cm and 15cm away.
- ☐ D The fan does not operate at all within this range.

CORRECT ANSWER ☒ C

2 What does a nested if statement allow programmers to do?

- ☐ A Check for multiple conditions sequentially.
- ☐ B Execute a single condition faster.
- ☐ C Ignore all conditions specified.
- ☐ D Only execute the outermost condition.

CORRECT ANSWER ☒ C

3 If a nested if statement in a MODI module setup checks first for motion and then for light, what modules are likely being used?

- ☐ A A) LED and Speaker.
- ☐ B Network and Dial.
- ☐ C ToF sensor and Environment sensor.
- ☐ D Button and LED.

CORRECT ANSWER ☒ C

4 What does the mathematical expression $10 < x \leq 20$ mean for the value of x?

- ☐ A x is greater than 10 and less than or equal to 20.
- ☐ B x is less than 10 and more than 20.
- ☐ C x is less than or equal to 10 and greater than 20.
- ☐ D x is exactly 10 or 20.

CORRECT ANSWER ☒ A

5 Which number fits within the range defined by $15 \leq y < 25$?

- ☐ A 15
- ☐ B 10
- ☐ C 25
- ☐ D 30

CORRECT ANSWER ☒ A

6 If a range is given as $5 \leq z < 15$, which of the following is NOT within this range?

- ☐ A 5
- ☐ B 10
- ☐ C 15
- ☐ D 14

CORRECT ANSWER ☒ C

7 What range does the expression $0 < w \leq 50$ represent?

- ☐ A w can be any number from 1 to 50, inclusive.
- ☐ B w can be any number from 0 to 50, but not including 0.
- ☐ C w can be any number from 0 to 51, excluding 51.
- ☐ D w must be exactly 0 or 50.

CORRECT ANSWER ☒ A

 Critical Thinking Challenge

- 1 Write down your idea on how understanding the operational range of devices like smart fans can help in designing energy-efficient homes.

- 2 Reflect on how using nested if statements in household appliances could improve safety. Write a paragraph discussing a specific example, such as a smart oven that turns off automatically.